

The First Logician: Aristotle's Categorical Syllogism

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GRADE LEVEL	DURATION	TOPICS
Grades 11-12 / Intro College	65 minutes	Logic

“What makes an argument's form alone — apart from what it's actually about — enough to guarantee that true premises lead to a true conclusion?”

Learning Objectives

By the end of this lesson, students will be able to:

- Identify the four categorical propositional forms (A, E, I, O) and translate ordinary statements into them.
- Construct a valid categorical syllogism in proper form (major premise, minor premise, conclusion) and identify its middle term.
- Distinguish valid categorical syllogisms from invalid ones by testing the argument's form independently of whether its conclusion happens to be true.

Materials Needed

- Handout listing the four categorical forms (All S are P / No S are P / Some S are P / Some S are not P) with labeled examples
- Set of syllogism cards, some valid and some invalid, mixing true and false content so form and truth can be pulled apart
- Whiteboard for diagramming the middle term across premises
- Venn diagram template handout for testing syllogism validity

Lesson Procedure

Hook: True Conclusion, Bad Argument

8 min

Present the class with an argument that has a true conclusion but a flawed structure, such as 'All fish live in water. All sharks live in water. Therefore, all sharks are fish.' Ask whether this is a good argument. Most students will initially say yes because the conclusion sounds plausible; guide them to notice the conclusion doesn't actually follow, previewing that today is about form, not content.

Direct Instruction: The Four Categorical Forms

15 min

Introduce Aristotle's *Organon* briefly as the founding texts of formal logic, then teach the four standard-form categorical propositions: A (universal affirmative), E (universal negative), I (particular affirmative), and O (particular negative). Have students translate several ordinary English sentences into these forms as a quick check.

Direct Instruction: Anatomy of the Syllogism

12 min

Explain the three-term, three-proposition structure of a categorical syllogism, and the role of the middle term, the term appearing in both premises but not in the conclusion, in linking the major and minor terms. Work through one classic valid example on the board and diagram the middle term visually.

Small-Group Practice: Sort the Syllogisms

20 min

In pairs or small groups, students receive a set of syllogism cards mixing valid and invalid, true-content and false-content combinations, and must sort them into valid versus invalid using Venn diagrams, explicitly separating whether an argument is valid from whether they happen to agree with the conclusion.

Closure: Build Your Own

10 min

Each student writes one original valid categorical syllogism and one original invalid one that looks tempting, about a topic of their choice, then trades with a partner who must identify which is which and explain why using the middle term.

Discussion Questions

- 1 Why did Aristotle think that the validity of an argument could be determined entirely by its form, without knowing anything about what the terms actually refer to?
- 2 Is it possible for a categorical syllogism to be valid but still be a bad argument in some other way? What would that require?
- 3 Why is the middle term's role in a syllogism essential — what would go wrong if a syllogism had four terms instead of three?
- 4 Can you think of an invalid syllogism whose conclusion happens to be true? What does that show about the relationship between validity and truth?
- 5 Aristotle's system covers only categorical statements about classes of things (all, no, some). What kinds of everyday reasoning does this system struggle to capture?

Assessment

Give students five new categorical syllogisms they haven't seen before, a mix of valid and invalid with both true and false content. Students must identify the middle term in each, diagram it, and determine validity, writing one sentence justifying each verdict. Grade for correct identification of validity based on form, independent of whether students happen to agree with the conclusions.



Extension Activity

Have students find or invent a syllogism used in a real advertisement, political speech, or op-ed, translate it into standard categorical form, identify the middle term, and evaluate whether it is actually valid, then write a short paragraph on whether the argument's persuasive power depends on its validity or on something else, like emotional appeal or ambiguous terms.

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